

Hongrui Wu

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EDUCATION

Stanford University

Stanford, CA | *Sep 2026 - Jun 2028*

M.S. in Electrical and Electronics Engineering
Computer Science Track (Artificial Intelligence Coursework)

Tongji University

Shanghai, China | *Sep 2021 - Jun 2026*

B.A. in Computer Science and Technology
Score: 91.27/100 **GPA:** 3.95/4.0 National Scholarship

SELECTED PUBLICATIONS

* Equal contribution. All listed papers are (co-)first-authored.

- [1] **Hongrui Wu***, K. Tuo*, C. Wang*, G. Li, S. Dong, X. Yu, H. Hou, Z. Peng, and J. Wang, “MIRA: A multimodal illustrated deep research agent for controllable reports,” Submitted to ICLR 2027, 2026.
- [2] K. Tuo*, C. Wang*, S. Dong*, **Hongrui Wu***, S. Wang, X. Yin, Y. Cao, N. Duan, and J. Wang, “MMMC: Towards realistic conversational multimodal instruction following,” Accepted to NeurIPS 2026.
- [3] X. Zhang*, S. Yoo*, **Hongrui Wu***, C. Li, J. Xie, and Z. Tu, “Pixarmesh: Autoregressive mesh-native single-view scene reconstruction,” in *CVPR*, 2026. arXiv: [2603.05888](#) [[cs.CV](#)].
- [4] **Hongrui Wu***, Z. Gao*, J. Cao, K. Yao, W. Shen, and Z. Wei, “Folk: Fast open-vocabulary 3d instance segmentation via label-guided knowledge distillation,” in *ICASSP*, 2026. arXiv: [2510.08849](#) [[cs.CV](#)].
- [5] J. Cao*, **Hongrui Wu***, Z. Feng, H. Bao, X. Zhou, and S. Peng, “Universe: Unleashing the scene prior of video diffusion models for robust radiance field reconstruction,” in *ICCV*, 2025. arXiv: [2510.01669](#) [[cs.CV](#)].

EXPERIENCE

JD Explore Academy [[1](#), [2](#)] · **Research Intern** · Mentor: **Jiaqi Wang**

May 2026 - Oct 2026

1. Multimodal DeepResearch Agent

- **MIRA-Harness** [[1](#)]: Designed a multimodal deep-research agent harness that turns a user’s report request, e.g. `[/concise-report]` or `[/detail-report]`, into checks the illustrated report must pass before delivery, at minimal extra cost; it adopts the `<tool_call>search/open/find/python</tool_call>` interface of most frontier models, natively compatible with them unlike task-specific open-source agents.
- **MIRA-Data**: Built an efficient, high-yield pipeline that synthesizes multi-topic, image–text interleaved search trajectories, training agents for efficient, long-horizon search with broader world knowledge; training on it made Qwen3.6-35B-A3B **SOTA** on **7** representative benchmarks among comparable work of similar parameter scale.
- **DeepResearchAgent-EvalKit** [[Code](#)]: We observed a key **Problem**: deep-research agents are evaluated inconsistently across the field, making results hard to compare and slow to obtain. **Solution**: a unified standard supporting past, present, and future text and multimodal benchmarks and diverse harness protocols; users just plug in their model or tool APIs. **Performance**: ~40% faster full evaluations that once took tens of hours, with consistent scores enabling fair comparison across models and benchmarks.

2. Multimodal Data Platform

- **JoyDataTool-Viewer** [[Demo](#)]: Tackled a daily **Problem**: agent, image–text interleaved, and text-heavy data is unreadable as raw JSON. **Solution**: a platform that renders each record by its task domain: images from paths, 3D boxes with coordinates for perception data, turn-by-turn tool calls for agent trajectories, plus eval verdicts and TensorBoard curves. **Performance**: heavily used team-wide across clusters.
- **JoyDataTool-Register** [[Demo](#)]: Addressed a scaling **Problem**: data from many contributors and domains must feed one large-scale training run in one format. **Solution**: a standardized intake pipeline: validation → tagging → field completion → pre-tokenization → dry run. **Performance**: one cross-cluster view of every dataset’s readiness and version, freeing stale copies and preventing mid-run SFT failures.

3. Multimodal Instruction-Following Benchmark [[2](#)]: Existing multimodal instruction-following benchmarks miss real multi-turn, image–video interleaved conversations. MMMC fills this gap with six formalized challenge types, a multi-agent, human-verified data pipeline, and rubric-guided training that improves constraint satisfaction without hurting

general ability. As co-first author, led evaluation of open-source, proprietary, and in-house models.

UC San Diego, MLPC Lab [3] · Research Intern

July 2025 - Dec 2025

CVPR 2026 Accepted [Paper] [Website] [Code] Advisor: Prof. Zhuowen Tu

Input: Single RGB Image Output: Scene-Level Native Meshes Model: EdgeRunner (0.8B)

- We develop the **first** scene-level 3D reconstruction framework that directly predicts **artist-created triangle meshes** from a single image, surpassing prior methods restricted to SDF-based geometry or object-level generation.
- We introduce a **unified autoregressive model** that jointly predicts instance poses, scales, and canonicalized geometry, enabling coherent multi-object reconstruction and spatially aligned scene assembly.
- Experiments on synthetic and real datasets show our method achieves **SOTA** reconstruction quality while producing lightweight, high-fidelity meshes suitable for downstream applications like robotics and other embodied systems.

Tongji University [4] · Research Intern

Nov 2024 - Aug 2025

ICASSP 2026 Accepted [Paper] Advisor: Prof. Zhihua Wei and Prof. Wen Shen

Input: Scene Point Clouds Output: Instance Masks & Class Labels Model: training-free

- FOLK constructs a **fast open-vocabulary 3D instance segmentation framework** that eliminates the 3D-to-2D projection mapping. By distilling open-vocabulary knowledge from 2D CLIP into a 3D student model, it removes noise from occlusions and computational redundancy, enabling direct semantic understanding.
- Proposes an innovative **knowledge distillation architecture**: The teacher network employs a **multi-view selection algorithm** combined with **density-guided mask completion** to generate high-quality 2D embeddings; the student network learns the 3D embedding space directly via a **label-guided distillation mechanism**, ensuring semantic consistency while achieving end-to-end efficient inference.
- Achieves SOTA performance with an **AP₅₀ = 35.7 on the ScanNet200 dataset**. Inference speed is accelerated by **6.0×–152×** compared to existing SOTA methods (single-scene latency as low as **3.64s**), attaining the highest matching rate with the fewest proposal candidates.

Zhejiang University [5] · Research Intern

Dec 2024 - Mar 2025

ICCV 2025 Accepted [Paper] [Website] Advisor: Prof. Sida Peng and Prof. Xiaowei Zhou

Input: Low-Quality Photos Output: Consistent Scene Reconstruction Model: DynamiCrafter (1.4B)

- UniVerse proposes a **robust reconstruction framework leveraging video diffusion model (VDM) scene priors**. It addresses inconsistencies in unstructured multi-view images caused by illumination variations and transient occlusions, restoring them into consistent static image sets to provide clean inputs for downstream reconstruction.
- We construct a **decoupled two-stage architecture separating "sequence restoration" from "geometric reconstruction"**: The first stage employs a multi-view sorting algorithm and zero-frame interpolation to build an initial video, then utilizes pre-trained VDM with semantic mask guidance and style control to repair inconsistent images into a geometrically consistent sequence. The second stage performs high-precision NeRF/3DGS reconstruction on the restored images, avoiding the instability of direct optimization on inconsistent data.
- Experiments on both synthetic and real datasets shows that our method significantly outperforms **SOTA** baselines, achieving superior results in PSNR, SSIM while effectively eliminating artifacts and floaters. UniVerse also supports flexible style control of the reconstructed 3D scene via reference images, showcasing exceptional generalization.

SKILLS

* Hands-on with all of these, not (yet) an expert in all.

Coding Agent Stack 🌸 Codex + 🌸 Claude Code | Power User (Personal Max, \$200/month)

ML / Systems PyTorch, Transformers, veRL, vLLM, Ray, Megatron-LM, LLaMA-Factory, LoRA